

Fractions, Decimals & Percents

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Fraction → decimal: divide top by bottom. Decimal → percent: multiply by 100 (move the decimal 2 places right). Percent → fraction: write it over 100, then simplify. Anchors: $\frac{1}{2} = 0.5 = 50\%$, $\frac{1}{4} = 0.25 = 25\%$.

Round 1 — Retrieve It

1. How do you change a fraction into a decimal?

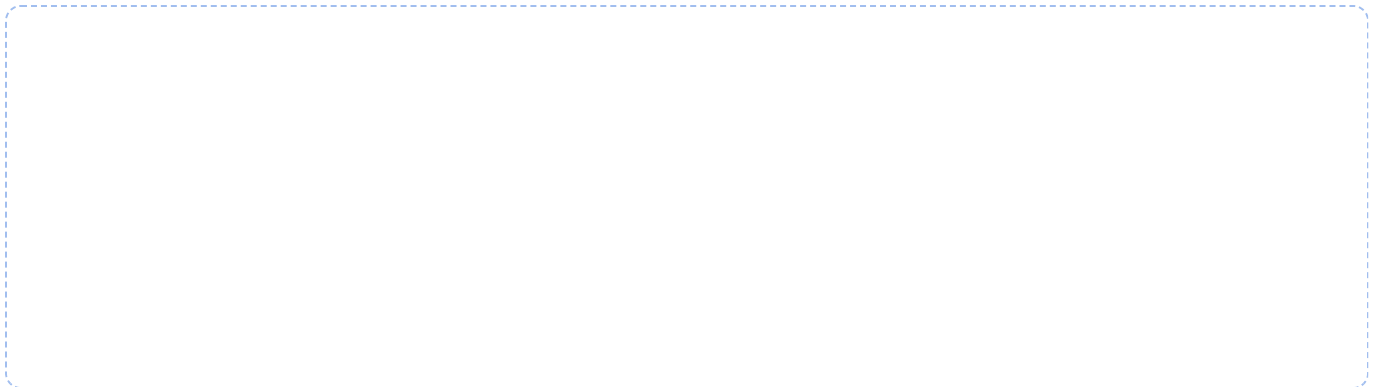
2. How do you change a decimal into a percent?

3. How do you change a percent into a fraction?

4. Write $\frac{1}{4}$ as a decimal and as a percent.

Round 2 — Sketch It

Draw a 10-by-10 grid (100 squares). Shade 50 squares. Label the shaded part as a fraction, a decimal, and a percent.



Fractions, Decimals & Percents — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Write $\frac{1}{2}$ as a decimal and as a percent.

2. Change 0.25 into a percent.

3. Change 50% into a simplified fraction.

Round 4 — Reason from Evidence

A student changes 0.25 to a percent and writes 2.5%. Explain what went wrong with moving the decimal, then give the correct percent.

Fractions, Decimals & Percents — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why $\frac{1}{2}$, 0.5, and 50% all represent the same amount. Use one example that is not printed on the cue card.

C**Connect**

Connect "divide top by bottom" to "over 100." How do these two moves let you travel between fractions, decimals, and percents in either direction?

E**Extend**

Create your own value and write it three ways: as a fraction, a decimal, and a percent. Show the work that connects them.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Fractions, Decimals & Percents

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Divide the numerator (top) by the denominator (bottom).
2. Multiply by 100, which moves the decimal point 2 places to the right.
3. Write the percent over 100, then simplify.
4. $1/4 = 0.25 = 25\%$.

Sketch It — look for

A 10×10 grid with 50 squares shaded, labeled as **50/100** (or $1/2$), **0.5**, and **50%**.

Apply It

1. $1/2 = 1 \div 2 = \mathbf{0.5} = \mathbf{50\%}$.
2. $0.25 \times 100 = \mathbf{25\%}$.
3. $50\% = 50/100 = \mathbf{1/2}$ (divide by GCF 50).

Reason from Evidence — look for

The decimal was moved the wrong way (or only one place). To make a percent you move the decimal 2 places right: $0.25 \rightarrow \mathbf{25\%}$, not 2.5%.

ACE — Extend look-for

Many answers possible. Check that the three forms match the same value and that the work correctly converts among fraction, decimal, and percent.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.